

Write a Prolog Program to print particular bird can fly or not. Incorporate required query

To write and execute a **Prolog program to determine whether a particular bird can fly or not** using suitable queries.

Algorithm

1. Start the program.
2. Define facts for birds that can fly and cannot fly.
3. Define the `can_fly/1` predicate to check whether a bird can fly.
4. Enter the name of the bird as a query.
5. Prolog searches the database for a matching fact.
6. If the bird has the property `can_fly`, display true; otherwise, display false.
7. Stop the program.

Program

Save as **6.pl**:

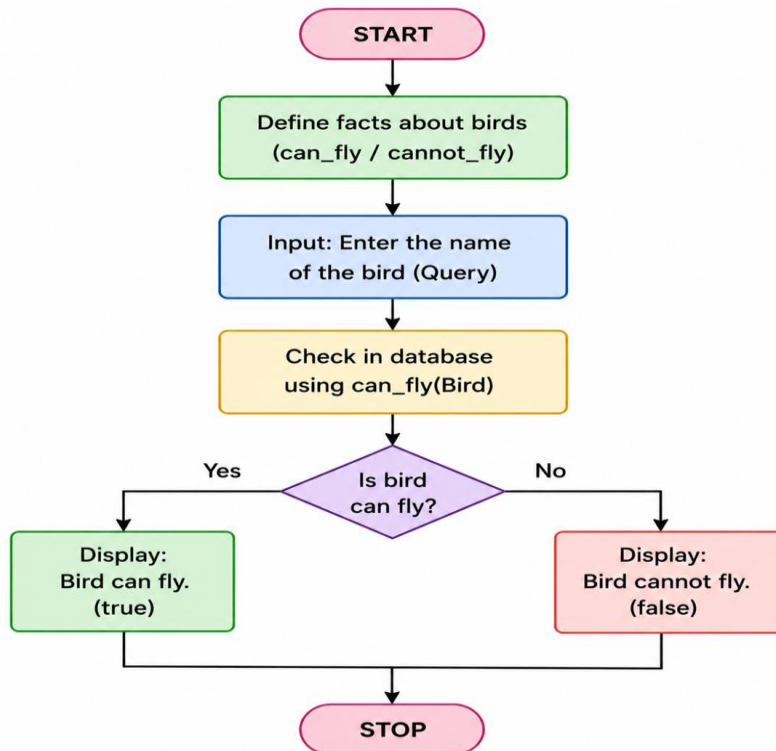
```
% Program to check whether a bird can fly or not
```

```
bird(ostrich, cannot_fly).  
bird(penguin, cannot_fly).  
bird(kiwi, cannot_fly).
```

```
bird(eagle, can_fly).  
bird(pigeon, can_fly).  
bird(sparrow, can_fly).
```

```
can_fly(Bird) :-  
    bird(Bird, can_fly).
```

Flow chart



Ouput

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Madhu Bavana/OneDrive/Desktop/Pr1/6.pl compiled 0.00 sec, 7 clauses
?-
| can_fly(eagle).
true.
?- can_fly(ostrich).
false.
?- bird(Bird, can_fly).
Bird = eagle .
?- can_fly(sparrow).
true.
?- can_fly(pigeon).
true.
?- bird(Bird, can_fly).
Bird = eagle
```

The screenshot shows the SWI-Prolog environment. The user has loaded a file '6.pl' which contains facts about birds and a predicate 'can_fly'. The output shows the results of several queries: 'can_fly(eagle)' is true, 'can_fly(ostrich)' is false, 'bird(Bird, can_fly)' returns 'Bird = eagle', 'can_fly(sparrow)' is true, and 'can_fly(pigeon)' is true. The final query 'bird(Bird, can_fly)' also returns 'Bird = eagle'.

Result

Hence Prolog Program to implement Towers of Hanoi is done